Yaswant Sai "Yash" Ejjagiri

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Education

University of Illinois at Urbana-Champaign Aug. 2021 – May 2025

Bachelor of Science in Computer Science + Astronomy, Minor in Physics and Math

Relevant Coursework: *Deep Learning for Computer Vision, Linear Algebra, Differential Equations, Statistics and Probability, Electromagnetic Fields, Classical Mechanics, Quantum Mechanics, Stellar Astrophysics, Computational Astrophysics, Applications of Machine Learning in Physics*

Research Experience

CRESST II Research Assistant Sep. 2026 – present

NASA Goddard Space Flight Center *Greenbelt, MD*

- Developing **neural-network emulators** of general-relativistic ray-tracing pulse-profile models for **NICER** X-ray data of PSR J0030+0451, replacing the millions of simulations each **Bayesian (MCMC)** fit needs to constrain neutron-star **mass**, **radius**, and **multipolar magnetic fields**
- Reproduced the bolometric haplo emulator workflow on **50M simulated light curves** with distributed **PyTorch** training on up to **64 A100 GPUs** (NASA HECC), and built a batched GPU pipeline benchmarking architectures on a common 100k-sample test set
- Extending emulators to the full **energy-resolved** signal by training on **50M 16-parameter phase–energy 2D maps**; diagnosing why bolometric loss fails to transfer to 2D and testing alternative **loss functions** and metrics against a **Poisson-noise baseline**

Research Assistant Sep. 2023 – Sep. 2026

National Center for Supercomputing Applications @ University of Illinois *Champaign, IL*

- Advised by Professor Xin Liu, working on Deep Learning-Driven (DeepDISC) **deblending** in astronomical imaging surveys using overlapping **space-based** (*Roman Space Telescope*) and **ground-based** (*Rubin/Large Synoptic Survey Telescope*) **coadds** | Project Repo
- Built and **parallelized** the joint Rubin–Roman **annotation pipeline** using multiprocessing and shared memory, generating MS-COCO-format labels with physically motivated matched-filter masks (using GalSim) for **100k cutouts containing 2M annotated galaxies and stars**, cutting total processing time by 99%
- Evaluated multiple Rubin–Roman fusion strategies including dual-encoder (Swin Transformer) **cross-modal contrastive learning**, knowledge distillation, and single-encoder channel concatenation; developed **embedding diagnostic tools** (cosine similarity, PaCMAP, feature ablation) to analyze learned representations across designs
- Built a custom **two-stage Friends-of-Friends and Hungarian catalog-matching pipeline** classifying detections into 9 blend and failure modes, plus a centralized metrics framework computing 15 scalar and binned metrics that are used across all downstream analysis
- Achieved a **10% increase in detection completeness and a 3–5% point reduction in unrecognized blend fraction** using the joint-survey concatenated model, directly mitigating systematic errors that degrade weak lensing measurements and photometric redshift inferences
- Mentored an undergraduate to build and deploy a DeepDISC-UX framework on cloud-based Rubin Science Platform, **architecting the Django/Jinja stack** and implementing the core `dd_lib` library for end-to-end workflows

Research Contributor Jun. 2026 – present

Summer of Open AI Research @ EleutherAI *Remote*

- Selected for a mentored open-science program in **interpretability** and **representation learning** extending UniverseTBD's Platonic Universe
- Designing a methodology, using **linear probing**, to test whether **astronomy foundation models** encode physical galaxy properties (mass, luminosity, SFR) as recoverable directions in their representations
- Extracted **layerwise embeddings** from every layer of ViT-base on ~**100k crossmatched HSC–Legacy Survey galaxy image pairs** for the alignment (MKNN, CKA, Procrustes) and physical-transfer analyses
- Co-authored *Shared Geometry Is Not Shared Physics* (NeurIPS 2026 workshop), a **layerwise test of the Platonic Representation Hypothesis** across 36 pretrained models and 4 crossmatched surveys, showing that models can share cross-survey geometry without sharing a transferable encoding of physical properties

Publications

- **Ejjagiri, Y.S.**, Merz, G., Liu, X. "DeepDISC: Deep Learning-Driven Deblending Using Overlapping Roman and Rubin LSST Data" *under DESC Internal Review*
- Yadav, P., **Ejjagiri, Y.S.**, Merz, G., Liu, X., "DeepDISC on LSST Data Preview 1: Scene-Level Detection, Deblending, and Star/Galaxy Classification in the ECDFS and EDFS Fields," *To be submitted to The Open Journal of Astrophysics*, arXiv:2610.10986 [astro-ph.IM]
- Merz, G., Liu, X., Schmidt, S., et al. (incl. **Ejjagiri, Y.S.**), "DeepDISC-Photoz: Deep Learning-Based Photometric Redshift Estimation for Rubin LSST," *The Open Journal of Astrophysics* (2025)
- Duraphe, K., Kannappan, A., et al. (incl. **Ejjagiri, Y.S.**), "Shared Geometry Is Not Shared Physics: A Layerwise Test of the Platonic Representation Hypothesis in Astronomy," *Accepted to NeurIPS 2026 Workshop: Interpretability for Discovery*, arXiv:2610.04130 [astro-ph.IM]
- Krishnan, A., **Ejjagiri, Y.S.**, Merz, G., Yadav, P., Liu, X. "DeepDISC Photometric Redshifts in LSST Data Preview 1: A Pixel-Level Normalizing-Flow Photo-z Estimator for the ECDFS Deep Field" *in preparation*

Presentations & Talks

Energy Resolved NN Emulators for Joint Inference of Neutron Star Properties with NICER

- Poster – AAS 249th Winter Meeting | *View Poster (TBA)* Jan. 2027
- Poster & Lightning Talk – JSI NICER Legacy Workshop | *View Poster & Talk* Nov. 2026

DeepDISC: Deep Learning-Driven Deblending Using Overlapping Roman and Rubin LSST Data

- Poster – AAS 247th Winter Meeting | *View Poster* Jan. 2026
- Poster & Lightning Talk – Dark Energy Science Collaboration Meeting | *View Poster & Talk* Jul. 2025
- Poster & Lightning Talk – Cosmic Cartography with Roman | *View Poster & Talk* Jul. 2025
- Poster & Lightning Talk – Center for Decoding the Universe Annual Conference | *View Poster & Talk* Jun. 2025

Teaching Experience

Guest Lecturer Apr. 2026

ASTR 596: AI and Big Data in Astronomy @ University of Illinois *Champaign, IL*

- Designed and delivered a **80-min hands-on lab** on **multimodal representation learning** for joint astronomical surveys, covering self-supervised pretraining across different data modalities (images, spectra, light curves) and cross-modal contrastive learning for shared embedding spaces
- Created all lab materials for **12 graduate students** including a Google Colab notebook with live coding exercises, a slide deck, and a comprehensive homework assignment

Undergraduate Grader Aug. 2024 – May 2025

Department of Astronomy @ University of Illinois *Champaign, IL*

- Graded weekly homeworks, labs, and discussion posts for **100+ students** providing written feedback across *ASTR 100: Introduction to Astronomy* and *ASTR 350: Big Bang, Black Holes, and the End of the Universe*

Outreach Experience

President Sep. 2021 – Sep. 2024

Astronomical Society @ University of Illinois *Champaign, IL*

- Supervised and coordinated Public Open Houses with **250+ attendees**, identifying visible planets through 12" Refractor Dome Telescope and educating guests on the history of the Observatory
- Redesigned club website with **NextJS** to increase functionality and accessibility for members and general public
- Built **1st-place exhibit** (*Stellar Solar Observing*) for ***Outstanding Freshman Exhibit*** at annual Engineering Open House 2024, engaging over **300+ guests** of all ages and backgrounds
- Facilitated **bi-weekly club meetings** aimed at educating members about astronomy through presentations and professional workshops featuring professors and graduate students as guest speakers

Work Experience

Technical Project Manager Jan. 2022 – May 2025

Introduction to Computer Science Honors @ University of Illinois *Champaign, IL*

- Mentored **25+ students** in developing NextJS + Supabase (PostgreSQL) web applications, React Native + Firebase applications, NodeJS applications with Express, and simulations using p5js and VPython

- Created **15+ effective and personalized tutorials** on building full-stack applications in programming languages such as NextJS, React, Unity, Python, JavaScript and HTML/CSS
- Established a per-team delivery process anchored by an **MVP specification document** and a **GitHub Kanban board**, with weekly check-ins driving teams toward fixed demo deadlines
- Built reusable **Next.js + Supabase starter templates** encapsulating authentication and database boilerplate, plus **cross-platform onboarding guides** for Windows and Mac dev environments, eliminating setup time
- Provided hands-on technical mentorship including **Postgres schema design from scratch** for multiple teams and debugging sessions resolving asynchronous data fetching errors and React state bugs

Full-Stack Web Developer & Computer Vision Engineer Jan. 2023 – Aug. 2023

Applied Technologies for Learning in the Arts & Sciences (ATLAS) @ University of Illinois *Champaign, IL*

- Trained a **YOLOv8 computer vision model** to detect the number of people visible in a YouTube live stream of the University of Illinois Quadrangle
- Annotated a custom dataset of **450 Quad Cam images** and trained different iterations of YOLOv8s, changing parameters such as image size, learning rate, and non-maximum suppression + IOU threshold
- Achieved a **50% improvement in model performance** (Mean Average Precision: 0.60 → 0.90) through hyperparameter tuning and by utilizing YOLOv8s-p2 to detect and localize smaller objects more accurately
- Deployed the model to a **Flask application** that performs **real-time inference** and presents detections through a periodically updated histogram to provide valuable marketing data for student organizations | Project Repo
- Developed a **full-stack web application** to generate customized paragraphs (Spotlights) about ATLAS Interns utilizing NodeJS, Express, EJS, CSS, and **OpenAI's** gpt-3.5-turbo **API** | Project Repo
- Implemented **OpenID Connect authentication** with Azure Active Directory through Passport.js
- Integrated **account management** with all CRUD operations using **MySQL and Sequelize** to store interns and their Spotlights, staff, and admin, streamlining the workflow of editing and publishing Spotlights

Skills

Languages: Python, C++, C, SQL (MySQL, PostgreSQL), JavaScript, HTML/CSS, R, MATLAB
Frameworks & Developer Tools: PyTorch, `Detectron2`, Astropy, GalSim, React, React Native, NextJS, Node.js, Express, Supabase, Firebase, Flask, Azure, Docker, AWS, Git, SLURM, Visual Studio Code